N83 Fixed-Base First-Fall Panel: 144 Systems

Candidate PDP V2 run report

2026-11-02

Observed outcome

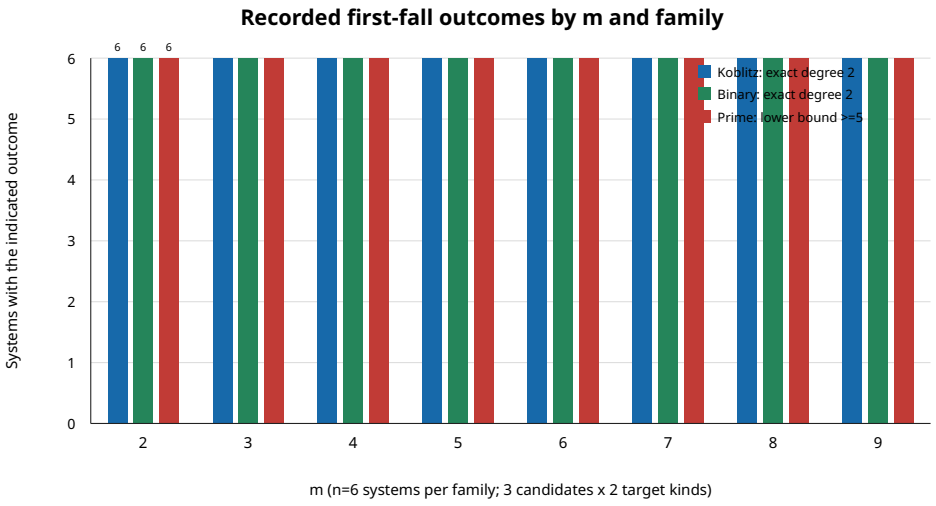
The frozen V2 first-fall procedure produced and freshly replayed certificates for all 144 pre-existing fixed-base systems across spike35–spike37, $m = 2$–9, Koblitz/binary/prime families, and planted/natural targets. The one certificate from the original attempt is preserved separately after its runner raised an exception; the continuation completed the other 143 cases. The independent analyzer reports 143/143 attempt-02 certificates hashed and bound, 143/143 fresh replays, and no problems. The recovered attempt-01 certificate has a successful independent fresh replay and matching S3 readback hash.

The first attempt's producer exit code and stdout/stderr were not captured before the replay-helper exception. That process outcome remains unknown; the certificate itself is present, hash-matched to S3, and replay-verified. The continuation recorded no failed or timed-out native cases.

Counts and outcomes

Family	Exact first-fall degree 2	Lower bound ≥ 5 (exact not found)
Koblitz	48	0
Binary	48	0
Prime	0	48
Total	96	48

Each family/m cell contains six systems (3 candidate IDs $\times$ 2 target kinds), for $m = 2, \ldots, 9$. The plot shows recorded counts per cell; it does not estimate a population parameter.



Protocol and execution

Protocol: N83-FACTOR-BASE-PDP-FFD-LFD-20261031-V2, frozen digest 50cd6f8213789de2dfd54a3b33a067ed87cc0def761b7c98769be3e19aee47a9. Inputs were the existing 144 fixed-base systems (fixed candidate x-domain 2^{27}), not regenerated. Each case was sequential, bounded to 60 seconds and 8 GiB, with a fresh-process certificate replay.

The continuation was launched with `python3`, script `research/n83_candidate_pdp_20261028/evidence/n83-factorbase-variants/candidate-fall-20261102/attempt-02/run_remaining.py`, and argument `--run`. Per-case commands are preserved in each `attempt.json`; they invoke the pinned release binary `fall run` and then `fall replay`. The attempt-02 runner and analyzer hashes are in attempt-02/script-binding.json. Pre-data source commit: 16c0ed20cb0f625367af79fde27f76600fa88bc6. Execution host: ip-172-31-26-169.us-east-2.compute.internal, Linux x86_64.

Across the 143 continuation cases, recorded producer process elapsed time sums to 314.742 s (median 1.502 s, range 0.740–27.215 s). Recorded replay process elapsed time sums to 89.001 s (median 0.285 s, range 0.006–3.220 s). These are sums of per-case wall-clock durations, not CPU time or end-to-end campaign wall time. Attempt-01's producer duration was not captured.

Interpretation and next step

These records establish first-fall outcomes only. In this procedure, Koblitz and binary cases have degree-2 first-fall certificates; prime cases certify no fall through degree 4 and thus retain a lower bound of 5. The measurements do not establish exact last fall degree, affine degree of regularity, solving degree, or a factor-base ranking. The next technical step is to implement and freeze the binary-closure procedure needed for exact LFD, then apply it to this same fixed-base cohort.

Evidence and custody

The combined machine-readable record and per-case evidence are in research/n83_candidate_pdp_20261028/evidence/n83-factorbase-variants/candidate-fall-20261102/. The 143-case attempt-02 independent validation record is attempt-02/independent-validation.json. The attempt-01 runner error is retained in attempt-01/runner-error.json. The complete archive is runs/n83-ffd-lfd-factorbase-panel-20261102/v2/combined-144/panel-complete.tar.gz; its SHA-256, size, and readback are in combined-144/S3-CUSTODY-complete.json. S3 attempt prefixes and the 286 byte-identical attempt-02 case-object readbacks are also recorded there.